16. Se define, para $A, B \in \mathbb{R}^{n\times n}$, $A \circ B$ como $(A \circ B)_{ij} = A_{ij}B_{ij}$ (el producto elemento a elemento). Demuestre que:

a) Si $D \in \mathbb{R}^{n\times n}$ y sus columnas son d_k, entonces $(DD^T) \circ B = \sum_{k=1}^{n} \operatorname{diag}(d_k)B\operatorname{diag}(d_k)$

b) Si A y B son simétricas definidas positivas, $A \circ B$ es simétrica definida positiva.

c) Si $f(x) = c_0 + c_1x + \ldots + c_dx^d$ con $c_i \geq 0\ \forall i$, $X \in \mathbb{R}^{n\times n}$ es simétrica semidefinida positiva y se define $Y = f(X)$ como:

$$Y_{ij} = f(X_{ij}),\ \ i, j = 1, \ldots, n$$

entonces Y es semidefinida positiva.

a) $D \in \mathbb{R}^{n\times n}$, $D = [d_1\ d_2 \ldots d_n]$

$$\operatorname{diag}(d_k) = \begin{bmatrix} d_k^1 & & & \\ & d_k^2 & & \\ & & \ddots & \\ & & & d_k^n \end{bmatrix}$$

$$DD^T = [d_1\ d_2 \ldots d_n] \begin{bmatrix} d_1^T \\ d_2^T \\ \vdots \\ d_n^T \end{bmatrix} =$$

$$= d_1 d_1^T + d_2 d_2^T + \ldots + d_n d_n^T$$

$$d_k d_k^T \begin{bmatrix} d_k^1 \\ \end{bmatrix}$$

$$d_k d_k^T = \begin{bmatrix} d_k^1 \\ d_k^2 \\ \vdots \\ d_k^n \end{bmatrix} \begin{bmatrix} d_k^1 & d_k^2 & \cdots & d_k^n \end{bmatrix} =$$

$$= \begin{bmatrix} (d_k^1)^2 & d_k^1 d_k^2 & \cdots & d_k^1 d_k^n \\ d_k^2 d_k^1 & (d_k^2)^2 & \cdots & d_k^2 d_k^n \\ \vdots & \vdots & \ddots & \vdots \\ d_k^n d_k^1 & d_k^n d_k^2 & \cdots & (d_k^n)^2 \end{bmatrix} \quad \forall k$$

$$(D D^T) \circ B = (d_1 d_1^T + d_2 d_2^T + \cdots + d_n d_n^T) \circ B =$$

Matriz

$$= d_1 d_1^T \circ B + d_2 d_2^T \circ B + \cdots + d_n d_n^T \circ B =$$

$$(A \circ B)_{ij} = A_{ij} B_{ij}$$

$$= \begin{bmatrix} (d_1^1)^2 b_{11} & d_1^1 d_1^2 b_{12} & \cdots & d_1^1 d_1^n b_{1n} \\ d_1^2 d_1^1 b_{21} & (d_1^2)^2 b_{22} & \cdots & d_1^2 d_1^n b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_1^n d_1^1 b_{n1} & d_1^n d_1^2 b_{n2} & \cdots & (d_1^n)^2 b_{nn} \end{bmatrix} +$$

$$+ \begin{bmatrix} (d_2^1)^2 b_{11} & d_2^1 d_2^2 b_{12} & \cdots & d_2^1 d_2^n b_{1n} \\ d_2^2 d_2^1 b_{21} & (d_2^2)^2 b_{22} & \cdots & d_2^2 d_2^n b_{2n} \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix} + \cdots +$$

$$\begin{bmatrix} \vdots & & & \\ d_2^n d_2^1 b_{n1} & d_2^n d_2^2 b_{n2} & \cdots & (d_2^n)^2 b_{nn} \end{bmatrix}$$

$$+ \begin{bmatrix} (d_n^1)^2 b_{11} & d_n^1 d_n^2 b_{12} & \cdots & d_n^1 d_n^n b_{1n} \\ d_n^2 d_n^1 b_{21} & (d_n^2)^2 b_{22} & \cdots & d_n^2 d_n^n b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_n^n d_n^1 b_{n1} & d_n^n d_n^2 b_{n2} & \cdots & (d_n^n)^2 b_{nn} \end{bmatrix} =$$

$$= \begin{bmatrix} d_1^1 b_{11} & d_1^1 b_{12} & \cdots & d_1^1 b_{1n} \\ d_1^2 b_{21} & d_1^2 b_{22} & \cdots & d_1^2 b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_1^n b_{n1} & d_1^n b_{n2} & \cdots & d_1^n b_{nn} \end{bmatrix} \begin{bmatrix} d_1^1 & & & \\ & d_1^2 & & \\ & & \ddots & \\ & & & d_1^n \end{bmatrix} +$$

$$+ \begin{bmatrix} d_2^1 b_{11} & d_2^1 b_{12} & \cdots & d^1 b_{1n} \\ d_2^2 b_{21} & d^2 b_{22} & \cdots & d_2^2 b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_2^n b_{n1} & d_2^n b_{n2} & \cdots & d_2^n b_{nn} \end{bmatrix} \begin{bmatrix} d_2^1 & & & \\ & d_2^2 & & \\ & & \ddots & \\ & & & d_2^n \end{bmatrix} + \cdots +$$

$$+ \begin{bmatrix} d_n^1 b_{11} & d_n^1 b_{12} & \cdots & d_n^1 b_{1n} \\ d_n^2 b_{21} & d_n^2 b_{22} & \cdots & d_n^2 b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_n^n b_{n1} & d_n^n b_{n2} & \cdots & d_n^n b_{nn} \end{bmatrix} \begin{bmatrix} d_n^1 & & & \\ & d_n^2 & & \\ & & \ddots & \\ & & & d_n^n \end{bmatrix} =$$

$$= \begin{bmatrix} d_1^1 & & & \\ & d_1^2 & & \\ & & \ddots & \\ & & & d_1^n \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix} \begin{bmatrix} d_1^1 & & & \\ & d_1^2 & & \\ & & \ddots & \\ & & & d_1^n \end{bmatrix} +$$

$$
+ \begin{bmatrix} d_2^1 & & & \\ & d_2^2 & & \\ & & \ddots & \\ & & & d_2^n \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix} \begin{bmatrix} d_2^1 & & & \\ & d_2^2 & & \\ & & \ddots & \\ & & & d_2^n \end{bmatrix} + \dots +
$$

$$
+ \begin{bmatrix} d_n^1 & & & \\ & d_n^2 & & \\ & & \ddots & \\ & & & d_n^n \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix} \begin{bmatrix} d_n^1 & & & \\ & d_n^2 & & \\ & & \ddots & \\ & & & d_n^n \end{bmatrix} =
$$

$$
= \operatorname{diag}(d_1) B \operatorname{diag}(d_1) +
$$

$$
+ \operatorname{diag}(d_2) B \operatorname{diag}(d_2) + \dots +
$$

$$
+ \operatorname{diag}(d_n) B \operatorname{diag}(d_n) =
$$

$$
= \sum_{k=1}^{n} \operatorname{diag}(d_k) B \operatorname{diag}(d_k)
$$

b) Si A y B son simétricas definidas positivas, $A \circ B$ es simétrica definida positiva.

A es simétrica y definida posi-
tiva $\Rightarrow \exists\, G \in \mathbb{R}^{n \times n}$, tr sup tal que

$A = G^T G$. Por conveniencia llamemos a $G = D^T$ y $G^T = D \Rightarrow A = D^T D$

Sea $x \neq 0$, $x \in \mathbb{R}^n$.

$$x^T(A \circ B)\,x \underset{A = DD^T}{=} x^T((DD^T)\circ B))\,x \overset{\text{Item a)}}{=}$$

$$= x^T\left(\sum_{k=1}^{n} \operatorname{diag}(d_k)\, B \operatorname{diag}(d_k)\right)x =$$

$$= \sum_{k=1}^{n} \underbrace{x^T \operatorname{diag}(d_k)}_{=y_k^T}\, B\, \underbrace{\operatorname{diag}(d_k)\, x}_{y_k^T} =$$

$$= \sum_{k=1}^{n} y_k^T B\, y_k > 0$$ pues B es definida positiva

Luego, $A \circ B$ es definida positiva

c) Si $f(x) = c_0 + c_1 x + \ldots + c_d x^d$ con $c_i \geq 0\ \forall i$, $X \in \mathbb{R}^{n\times n}$ es simétrica semidefinida positiva y se define $Y = f(X)$ como:

$$Y_{ij} = f(X_{ij}),\ \ i,j = 1,\ldots,n$$

entonces Y es semidefinida positiva.

$$Y = f(X) \underset{\substack{\uparrow \\ Y_{ij} = f(X_{ij}) \ \forall i j}}{=} \begin{bmatrix} c_0 + c_1 x_{11} + c_2 x_{11}^2 + \ldots + c_d x_{11}^d & c_0 + c_1 x_{12} + c_2 x_{12}^2 + \ldots + c_d x_{12}^d & \cdots & c_0 + c_1 x_{1n} + c_2 x_{1n}^2 + \ldots + c_d x_{1n}^d \\ c_0 + c_1 x_{21} + c_2 x_{21}^2 + \ldots + c_d x_{21}^d & c_0 + c_1 x_{22} + c_2 x_{22}^2 + \ldots + c_d x_{22}^d & \cdots & c_0 + c_1 x_{2n} + c_2 x_{2n}^2 + \ldots + c_d x_{2n}^d \\ \vdots & \vdots & \ddots & \vdots \\ c_0 + c_1 x_{n1} + c_2 x_{n1}^2 + \ldots + c_d x_{n1}^d & c_0 + c_1 x_{n2} + c_2 x_{n2}^2 + \ldots + c_d x_{n2}^d & \cdots & c_0 - c_1 x_{nn} + c_2 x_{nn}^2 + \ldots + c_d x_{nn}^d \end{bmatrix} =$$

$$= C_0 + c_0 X + c_1 X \circ X + \ldots + \underbrace{X \circ \ldots \circ X}_{d\text{-veces}}$$

donde,

$$C_0 = \begin{bmatrix} c_0 & c_0 & \cdots & c_0 \\ c_0 & c_0 & \cdots & c_0 \\ \vdots & \vdots & \ddots & \vdots \\ c_0 & c_0 & \cdots & c_0 \end{bmatrix}$$

Hayamos el siguiente producto:

$$X \circ Y = X \circ f(X)$$

$$= X \circ (C_0 + c_1 X + c_2 X \circ X + \ldots + c_d X \circ \ldots \circ X)$$

$$= C_0 X + c_1 X \circ X + c_2 X \circ X \circ X + \ldots + c_d \underbrace{X \circ \ldots \circ X}_{d+1\text{-veces}}$$

$\Rightarrow$ Sea $z \neq 0$, $z \in \mathbb{R}^n$

$$z^T (X \circ Y) z = z^T (C_0 X + c_1 X \circ X + c_2 X \circ X \circ X + \ldots + c_d X \circ \ldots \circ X) z =$$

$$= \underbrace{z^T C_0 X z}_{\geq 0} + c_1 z^T X \circ X z + c_2 z^T X \circ X \circ X z + \ldots + c_d z^T X \circ \ldots \circ X z$$

Luego, por item b) $X \circ Y$ es semidefinida positiva

Por otra parte,

como X es simétrica semidefinida finida $\Rightarrow \exists$ al menor una matriz D tr Sup tal que $X = DD^T$

(Imp: D no es única)

$$\Rightarrow 0 \leq z^T(X \circ Y)z$$

$$= z^T((DD^T) \circ Y)z$$

$$\underset{\uparrow}{=} z^T\left(\sum_{k=1}^{n} \operatorname{diag}(d_k)\, Y \operatorname{diag}(d_k)\right)z$$

Item a)

$$= \sum_{k=1}^{n} \underbrace{z^T \operatorname{diag}(d_k)}_{v_k}\, Y\, \underbrace{\operatorname{diag}(d_k) z}_{v_k}$$

$$= \sum_{k=1}^{n} v_k Y v_k \geq 0$$

$\Rightarrow Y$ es semidefinida positiva